

Experiment No.: 5

- Title:** Apply array methods and object handling; create a cart total calculator with discount logic.
- Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
- Theory:** The Shopping Cart Total Calculator is a web application developed using HTML, CSS, and JavaScript. It demonstrates the use of arrays and objects by storing multiple products as objects inside an array. The application accepts the product name, price, and quantity from the user, calculates the total amount using the `reduce()` array method, and applies a dynamic discount based on the total purchase amount. It also displays the final payable amount through a simple, attractive, and user-friendly interface.
- Output :**

le C:/Users/vithi/OneDrive/Desktop/flexi.py/vithिकासोजी%20javascript/P5.html

Smart Cart

VITHIKA SAOJI

✨ Add items and watch the discount grow automatically ✨

Add Item

Item NamePrice (₹)Quantity

e.g. Headphones01

+ Add to Cart

Your Cart

ITEM	PRICE	QTY	SUBTOTAL	
Laptop	₹70000.00	1	₹70000.00	<button>✕ Remove</button>
Headphones	₹1500.00	1	₹1500.00	<button>✕ Remove</button>

Bill Summary

TOTAL AMOUNT
₹71500.00

DISCOUNT (20%)
₹14300.00

FINAL AMOUNT PAYABLE
₹57200.00

Discount Tiers • Above ₹5000: 20% | Above ₹2000: 15% | Above ₹1000: 10% | Above ₹500: 5% | Below ₹500: 0%

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Smart Cart
VITHIKA SAOJI
 ✨ Add items and watch the discount grow automatically ✨

Add Item

Item Name:
 Price (₹):
 Quantity:
+ Add to Cart

Your Cart

ITEM	PRICE	QTY	SUBTOTAL	
Phone charger	₹100.00	1	₹100.00	<button>✕ Remove</button>
mouse	₹450.00	2	₹900.00	<button>✕ Remove</button>

Bill Summary

TOTAL AMOUNT
₹1000.00

DISCOUNT (5%)
₹50.00

FINAL AMOUNT PAYABLE
₹950.00

5. Result/Conclusion: The Shopping Cart application was successfully developed and executed. It accepts product details from the user, stores them using an array of objects, calculates the total amount using the `reduce()` method, and applies dynamic discounts based on the purchase amount. The application accurately displays the cart details, discount, and final payable amount. This project successfully demonstrates the practical implementation of array methods, object handling, user input, and discount calculation in JavaScript while providing a simple and interactive shopping cart interface.

Case Study No.: 05

1. Case Study Title: To study the functioning of Arrays and Loops in JavaScript

- Create and manipulate arrays using push(), pop(), shift(), unshift(), splice(), and slice().
- Use array methods map(), filter(), reduce(), and forEach().

Case Study: Create an array of numbers and find the maximum and minimum values.

2. Case Study Program Code:

```
P5.html x P5 case study.html x P6 email validation.html x
P5 case study.html > html > body > script > shiftElement
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Array Operations in JavaScript</title>
5
6 <style>
7     body {
8         background-color: lightblue;
9         font-family: Arial, sans-serif;
10        text-align: center;
11        padding: 40px;
12    }
13
14    .container {
15        background-color: white;
16        width: 600px;
17        margin: auto;
18        padding: 30px;
19        border-radius: 15px;
20    }
21
22    input {
23        padding: 10px;
24        width: 400px;
25        margin: 10px;
26        font-size: 16px;
27    }
28
29    button {
30        padding: 10px 18px;
31        margin: 5px;
32        border: none;
33        border-radius: 5px;
34        background-color: darkblue;
35        color: white;
36        cursor: pointer;
37    }
38
39    button:hover {
40        background-color: navy;
41    }
42
43    #output {
44        margin-top: 20px;
45        font-size: 18px;
46        text-align: left;
47        padding: 15px;
48    }
49 </style>
50 </head>
```

```

52 <body>
53
54 <div class="container">
55
56   <h2>Array Creation and Manipulation</h2>
57   <p>Enter numbers separated by commas:</p>
58
59   <input type="text" id="numbers"
60     placeholder="Example: 25, 10, 95, 5, 30, 15">
61
62   <br>
63   <button onclick="createArray()">Create Array</button>
64   <br><br>
65
66   <button onclick="pushElement()">Push</button>
67   <button onclick="popElement()">Pop</button>
68   <button onclick="shiftElement()">Shift</button>
69   <button onclick="unshiftElement()">Unshift</button>
70   <button onclick="spliceElement()">Splice</button>
71   <button onclick="sliceElement()">Slice</button>
72
73   <br><br>
74   <button onclick="findMaxMin()">Find Maximum & Minimum</button>
75   <div id="output"></div>
76
77 </div>
78
79 <script>
80
81   let numbers = [];
82
83   // Create Array
84   function createArray() {
85
86     let input = document.getElementById("numbers").value;
87     numbers = input.split(",").map(Number);
88     display("Array Created: " + numbers.join(", "));
89   }
90
91   // push() - adds element at the end
92   function pushElement() {
93     numbers.push(100);
94     display("After push(100): " + numbers.join(", "));
95   }
96
97   // pop() - removes last element
98   function popElement() {

```

```

P4 casestudy.html X P5.html X P5 case study.html
2 <html>
52 <body>
79 <script>
98   function popElement() {
99     numbers.pop();
100    display("After pop(): " + numbers.join(", "));
101  }
102  // shift() - removes first element
103  function shiftElement() {
104    numbers.shift();
105    display("After shift(): " + numbers.join(", "));
106  }
107  // unshift() - adds element at beginning
108  function unshiftElement() {
109    numbers.unshift(50);
110    display("After unshift(50): " + numbers.join(", "));
111  }
112  // splice() - removes/replaces elements
113  function spliceElement() {
114    numbers.splice(1, 1);
115    display("After splice(1, 1): " + numbers.join(", "));
116  }
117  // slice() - creates a portion of array
118  function sliceElement() {
119    let newArray = numbers.slice(1, 4);
120    display(
121      "Original Array: " + numbers.join(", ") +
122      "<br>After slice(1, 4): " + newArray.join(", ")
123    );
124  }
125  // Case Study - Find Maximum and Minimum
126  function findMaxMin() {
127    if (numbers.length === 0) {
128      display("Please create an array first.");
129      return;
130    }
131    let maximum = numbers[0];
132    let minimum = numbers[0];
133    for (let i = 1; i < numbers.length; i++) {
134      if (numbers[i] > maximum) {
135        maximum = numbers[i];
136      }
137      if (numbers[i] < minimum) {
138        minimum = numbers[i];
139      }
140    }
141    display(
142      "<b>Case Study Result</b><br><br>" +
143      "Array: " + numbers.join(", ") +
144      "<br>Maximum Value: " + maximum +

```

```

P4 casestudy.html X P5.html X P5 case study.html
P5 case study.html > html > body > script > findMaxMin
2 <html>
52 <body>
79 <script>
126   function findMaxMin() {
144     "<br>Maximum Value: " + maximum +
145     "<br>Minimum Value: " + minimum
146   };
147   }
148   // Display output
149   function display(message) {
150
151     document.getElementById("output").innerHTML = message;
152   }
153
154 </script>
155 </body>
156 </html>

```

3. Output:

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Array Creation and Manipulation

Enter numbers separated by commas:

Create Array

Push

Pop

Shift

Unshift

Splice

Slice

Find Maximum & Minimum

Array Created: 12, 23, 45, 89, 97

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Array Creation and Manipulation

Enter numbers separated by commas:

Create Array

Push

Pop

Shift

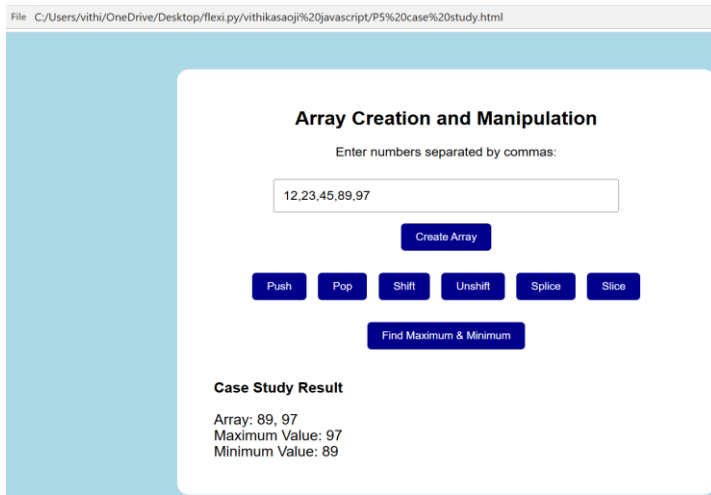
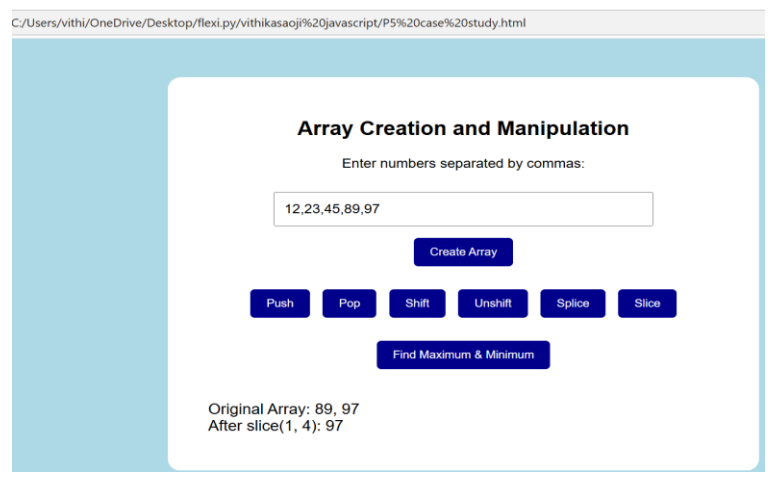
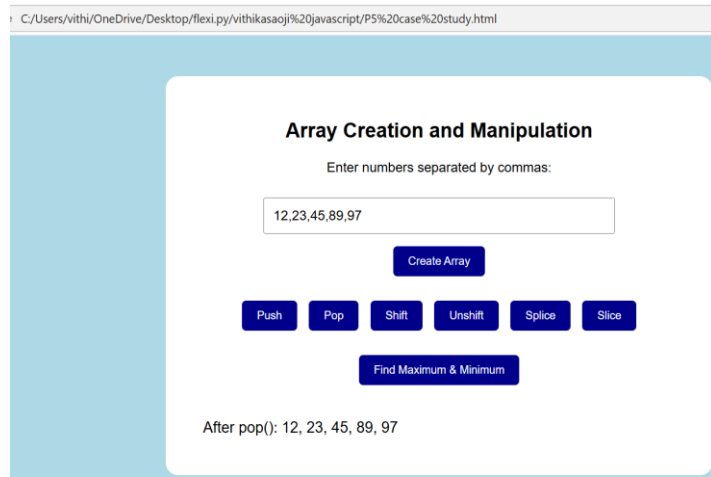
Unshift

Splice

Slice

Find Maximum & Minimum

After push(100): 12, 23, 45, 89, 97, 100



4. Result/Conclusion:

Thus, the functioning of JavaScript arrays and loops was studied and implemented successfully. Various array methods such as push(), pop(), shift(), unshift(), splice(), slice(), map(), filter(), reduce(), and forEach() were used to manipulate and process array elements. The maximum and minimum values were also successfully found from an array of numbers.